✦ E-COMMERCE PROJECT · STAGE P1 PROPOSAL

VELARIS.

Curated contemporary streetwear & sustainable capsule wardrobe essentials.

FRONTEND & UX LEAD

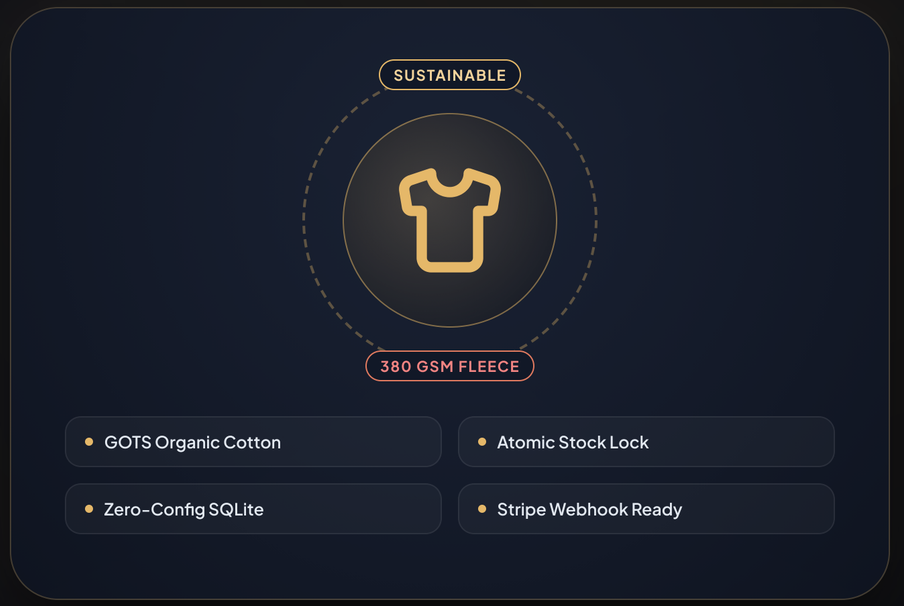
Predut Andreea-Denisa

Catalog, Cart, Checkout & Mobile UI

BACKEND & SYSTEMS LEAD

Sapca Roman-Stefan

Core APIs, Auth, Payments & Concurrency



MARKET FOUNDATION

Context & Value Proposition

Target Customer Personas

Focused on Gen-Z and young urban professionals (18–35) seeking durable, interchangeable capsule apparel with certified material provenance.

- **The Capsule Wardrobe Builder**
  Seeks high-versatility minimalist staples (boxy tees, heavyweight hoodies) designed to mix and match effortlessly without buying into micro-trends.
- **The Quality-Conscious Shopper**
  Demands transparent fabric specifications (GSM fabric weight, weave details) and clear garment sizing dimensions to prevent avoidable sizing returns.
- **The Mobile-First Digital Native**
  Expects instantaneous page loads, zero checkout obstacles, saved delivery profiles, and real-time order visibility.

Unique Value Proposition

Eliminating fast-fashion clutter through curated seasonal capsule drops, transparent sizing matrices, and guaranteed stock concurrency.

- **Curated Capsule Collections**
  Strictly seeded 30+ items across 4 structured categories: Heavyweight Hoodies, Boxy Tees, Structured Denim, and Weatherproof Outerwear.
- **Garment Transparency & Sizing Matrix**
  Every product page highlights fabric GSM density, care guides, and exact garment dimensions to foster buyer confidence.
- **Frictionless Checkout & Stock Truth**
  Authenticated cart persistence across sessions, multi-tier delivery options, and atomic inventory reduction strictly upon Stripe webhook confirmation.

COMPETITIVE INTELLIGENCE

Market Benchmark & Differentiators

Zalando

EUROPEAN RETAILER

BENCHMARK TO ADOPT

Rich multi-attribute category taxonomy (Hoodies, Tees, Denim, Outerwear) with responsive multi-angle photography and fluid mobile layouts.

OUR COMPETITIVE EDGE

Eliminate overwhelming thousands-of-items catalog bloat and dubious 3rd-party marketplace noise; provide a tightly curated capsule collection with guaranteed stock integrity.

Zara

HIGH-STREET FLAGSHIP

BENCHMARK TO ADOPT

High-fashion editorial lookbook visuals, elevated typography, and a streamlined multi-step order summary sequence.

OUR COMPETITIVE EDGE

Fix Zara's notoriously counter-intuitive, maze-like layout navigation; deliver instantaneous keyword search with clear zero-results feedback and reliable cart persistence across logins.

Farfetch

LUXURY MARKETPLACE

BENCHMARK TO ADOPT

Granular garment fit specs, transparent fabric composition breakdown, and clear multi-tiered courier delivery selections with calculated shipping costs.

OUR COMPETITIVE EDGE

Remove invasive pop-up cookie overlays and aggressive promo spam; deliver transparent server-validated discount rules and atomic inventory decrement at checkout.

SYSTEM ENGINEERING

Technology Stack & Justifications

PRESENTATION LAYER

ASP.NET Razor & Tailwind CSS

We chose Razor Views paired with Tailwind CSS to build responsive UI layouts down to 360px without requiring a detached frontend build pipeline.

- Mobile-first down to 360px (A11)
- Server-rendered instant loading
- Zero Node build dependency

APPLICATION CORE

ASP.NET Core (C#)

We selected ASP.NET Core because it provides high-performance custom API handling and built-in CSRF/security middleware.

- Server-side CSRF & Auth (A10)
- Admin Role Policies (A9)
- Async webhook processing (A6)

PERSISTENCE LAYER

SQLite & EF Core

SQLite managed via EF Core ensures zero-configuration local portability and reliable ORM data integrity for products and orders.

- Automated 30+ item seed script
- Strict relational foreign keys
- Optimistic Concurrency (A7)

TRANSACTION ENGINE

Stripe (Test Mode)

We selected Stripe's official .NET SDK to handle test card checkouts and webhook-based payment confirmation seamlessly.

- Local test via Stripe CLI (A6)
- Declined card test coverage
- Cryptographic webhook verify

MILESTONE BREAKDOWN

Execution Roadmap · Weeks 1–7 (P1 → P2)

SPRINT 1 · FOUNDATION

Weeks 1–3

Initialize GitHub repository, establish branching policies, model entity schemas, and deliver Presentation 1.

Joint Team Effort SETUP

Git repository initialization, README setup, domain modeling & P1 slides.

SPRINT 2 · DATA & CORE

Weeks 4–5

Database seeding (30+ garments across 4 categories: Hoodies, Tees, Denim, Jackets) and base responsive layouts.

Predut Andreea-Denisa:

A1 Product catalog, A2 Basic search, A11 Responsive design (360px).

Sapca Roman-Stefan:

A4 User accounts (Argon2/bcrypt), A10 Basic security & CSRF.

SPRINT 3 · PROTOTYPE

Weeks 6–7

Compile 8–page specification PDF (ERD, architecture, API routes, payment flow), build live catalog prototype, and deliver P2.

Joint Milestone P2 APPROVAL

Live working prototype (DB-driven catalog + login demo), commit tagged P2.

MILESTONE BREAKDOWN

Execution Roadmap · Weeks 8–12 (P2 → P3)

SPRINT 4 · COMMERCE

Weeks 8–9

Implement complete shopping cart state machine, order review history, and custom administration back-office.

Predut Andreea-Denisa:
A3 Shopping cart (server total, stock lock, auth persistence), A8 Order history.

Sapca Roman-Stefan:
A9 Administration panel (CRUD products/categories with image upload, order workflow).

SPRINT 5 · PAYMENTS & TESTS

Weeks 10–11

Multi-step checkout UX, asynchronous Stripe webhook processing, inventory concurrency protection, and automated testing.

Predut Andreea-Denisa:
A5 Checkout flow (shipping address, 2 carrier options, order confirmation page).

Sapca Roman-Stefan:
A6 Sandbox payment (webhook), A7 Stock management, A12 Automated tests (≥10 tests).

SPRINT 6 · HARDENING & P3

Week 12

Application hardening, bonus feature integration (B1 Filtered search, B5 Discount codes, B7 Wishlist), and final P3 live defense.

Predut Andreea-Denisa:
Bonus B1 (Filtered search in URL), Bonus B7 (Wishlist), UI mobile polishing.

Sapca Roman-Stefan:
Bonus B5 (Discount codes engine), Docker Compose script, P3 Live Demo rehearsals.

SYSTEM RESILIENCE

Risk Management & Mitigation Protocols

CRITICAL FLOW · A6

Webhook Failures & Order Desync

If asynchronous Stripe webhooks fail to reach our server due to local network drops or timeouts, orders remain stuck in *Unpaid* status even though the customer's test card was charged.

MITIGATION PROTOCOL

Extensively test webhook forwarding locally using the official Stripe CLI (`stripe listen --forward-to`), verify cryptographically signed webhook signatures, and enforce idempotent order state handlers.

DATA INTEGRITY · A7

Multi-Size Stock Overselling

Two concurrent shoppers attempting to purchase the last available unit of a popular garment size (e.g. Size M Hoodie) at the exact same second, causing negative inventory or order fulfillment conflicts.

MITIGATION PROTOCOL

Implement atomic database transactions and EF Core optimistic concurrency tokens (`RowVersion`) during payment confirmation to ensure over-subscribed checkouts are safely aborted.

PROJECT TIMELINE · A1–A12

Scope Creep & Apparel Complexity

Over-engineering complex matrix variations (multi-size, multi-color stock trees, custom measurement engines) early in development before securing the full 70 points for core requirements A1–A12.

MITIGATION PROTOCOL

Enforce a strict single-SKU/variant schema to guarantee 100% completion of requirements A1–A12 first; defer bonus features (B1, B5, B7) strictly to Week 12 once core points are verified.